



CEO decision horizon and firm performance: An empirical investigation

Murad Antia^{a,1}, Christos Pantzalis^{a,*}, Jung Chul Park^{b,2}

^a Department of Finance, College of Business Administration, BSN 3403, University of South Florida, Tampa, FL 33620, United States

^b Department of Economics and Finance, College of Business, P.O. Box 10318, Louisiana Tech University, Ruston, LA 71272, United States

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ABSTRACT

We investigate the effect of top managers' myopia on firms' market valuation. We devise a measure of expected CEO tenure as a proxy for the length of CEO decision horizon. After accounting for the endogenous nature of CEO horizon, our empirical tests show that shorter CEO horizon is associated with more agency costs, lower firm valuation and higher levels of information risk. The results are consistent with the notion that a short CEO decision horizon is indicative of preference for investments that offer relatively faster paybacks at the expense of long-term value creation.

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1. Introduction

Theoretical studies have shown that managerial opportunism is a key determinant of economic "short-termism". Moral hazard models with informational asymmetries indicate that managers prefer and pursue short-term results that serve their own interests at the expense of long-run results that would be optimal for their firms. In other words, decisions that may represent optimal personal intertemporal choices from a managerial perspective, may be suboptimal intertemporal choices from a firm perspective.³ Although, to our knowledge, there is no study that has directly tested the contention that managerial opportunism leads to a firm's failure to invest for the long run, there are several studies that have linked the propensity to forgo long-run investments with managers' limited tenure.⁴ In the past 20 years average CEO tenures have decreased from about 8 to less than 4 years (Le Breton-Miller and Miller, 2006) and, consequently, the pressure on CEOs to deliver quick results has increased

* Corresponding author. Tel.: +1 813 974 3262; fax: +1 813 974 3084.

E-mail addresses: mantia@coba.usf.edu (M. Antia), cpantzal@coba.usf.edu (C. Pantzalis), jcpark@latech.edu (J.C. Park).

¹ Tel.: +1 813 974 6350; fax: +1 813 974 3084.

² Tel.: +1 318 257 3571; fax: +1 318 257 4253.

³ Narayanan (1985) and Holmstrom and Ricart i Costa (1986) show that managers with private information about a project, attempting to enhance their reputations, have an incentive to accelerate the project's returns to the detriment of long-run performance. Hirshleifer and Thakor (1992) make a case for managers' preoccupation with safety, which leads them to prefer more short-term payoffs that have lower returns for the firm. Campbell and Marino (1994) maintain that managers select myopic investments because they want to indicate their superior abilities to labor markets, hoping they can exit the firm before their poor choices become apparent.

⁴ For example, Dechow and Sloan (1991) show that CEOs tend to decrease the R&D and advertising expenditures in their final years in office. There is also evidence that older CEOs are more likely to cut back on capital expenditures (Conyon and Florou, 2003). Bizjak et al. (1993) do not find any evidence that firms with CEOs near retirement are foregoing long-term investments. Mannix and Loewenstein (1994) using a laboratory experiment reach the opposite conclusion. They show that when CEOs expect to exit the firm they tend to focus more on short-term payoffs.

dramatically. This may explain why there have been so many acquisitions aimed at increasing revenues (Morck et al., 1990) and cost-cutting measures aimed at increasing profits. Furthermore, long-term investments that do not produce profits in the short-term are avoided, as CEOs tend to view these as being only beneficial for their successors (James, 1999).

One of the three main sources⁵ of conflict between managers and firms' stakeholders is that managers' decision horizons are shorter than shareholders' investment horizons (Jensen and Smith, 1985). Managers' claim on the firm is limited to their tenure, while a firm's lifespan is much longer. Since managerial decision horizons are limited to their expected tenure, managers approaching retirement age or about to be replaced become more “myopic,” in the sense that they tend to place less weight on cash flows occurring after their employment time horizon. A consequence of managerial myopia is that some good projects may be avoided.⁶

In this study we focus on the top manager of the firm, i.e. the CEO, and provide an empirical investigation of the relationship between CEO decision horizon and firm performance. There is a clear link between CEO age and the horizon problem, as supported by evidence that the turnover rate for CEOs increases with age (Ocasio, 1994). In addition, we argue that the horizon problem is more severe when a CEO's tenure with his/her firm is longer than usual, because longer CEO tenure increases commitment to the status quo and decreases willingness to adopt newer, improved techniques (Hambrick et al., 1993). Based on these insights, we use both CEO age and the length of current tenure to develop a relative measure of expected tenure, which we use to proxy the length of CEO decision horizon. Our empirical investigation entails three steps. First, we examine whether agency costs are greater if a firm's CEO has a short decision horizon. In the second step, we examine whether firm-specific information risk is greater for firms with shorter CEO decision horizon. Finally, we test whether expected tenure is significantly related with market valuation. Based on the premise that a CEO's expected tenure is a good approximation of his/her decision horizon, we hypothesize that firms should benefit when CEOs are expected to have a long tenure with their company. The benefits from longer expected tenure should be lower agency costs and information risk, and higher market valuation.

Our empirical findings provide support for all three hypotheses. We define agency costs of free cash flow as the occurrence of large amounts of cash when future growth opportunities are poor and find that firms with longer CEO decision horizon suffer from substantially lower agency costs than those with shorter CEO decision horizon. In addition we show that expected tenure is a significant determinant of both accruals quality and the probability of information-based trading (PIN) of Easley et al. (1996a, 1996b, 1997a, 1997b, 2002, 2005). As discussed in Francis et al. (2005) these are alternative measures of information risk. Accruals quality is found to increase with expected tenure, consistent with the notion that myopic CEOs are more likely to manipulate earnings. Firms with CEOs that have short expected tenure have greater PIN. Given the fact that information risk is greater when the uncertainty around the next few periods' cash flows increases,⁷ this result is consistent with the notion that CEO myopia also increases uninformed investors' risk of holding the stock, because the informational advantage of privately informed investors increases.⁸ Finally, we show that CEO expected tenure has a significant effect, both in statistical and in economic terms, on several measures of market valuation. Firms with managers who are relatively old or have held their positions for long time relative to their peers are more likely to be valued lower than firms with younger CEOs and/or CEOs that have not served on their current jobs for too long. Our findings are consistent with the view that longer expected tenure leads to an alignment of the interests of managers and shareholders, and, consequently to higher valuation. In sum, we show that CEO myopia diminishes firm performance.

In addition to providing direct evidence on the impact of decision horizon on firm performance, our study also contributes to the literature by providing supplemental evidence to recent findings of the effect of CEO compensation structure on market valuation and information risk. Our results show that CEO incentive compensation, consisting of unexercised stock options, is negatively related to his/her decision horizon. This evidence, in conjunction with the negative effect of the predicted value of CEO decision horizon on market valuation along with its positive impact on information risk and agency costs, suggests that excessive stock options rewards might decrease CEO decision horizon and thus lead to poor firm performance. Therefore, this study highlights an important channel (i.e., decision horizon) through which poorly designed executive compensation structure can affect firm performance.

The rest of the paper is organized as follows. In the next section, we review the related literature and develop the hypothesis. Section 3 describes the data selection process and introduces the measure of CEO's decision horizon. Sections 4 and 5 present the results of univariate tests and two-stage regressions, respectively. Finally, Section 6 summarizes and provides concluding remarks.

⁵ The other two sources of conflict between managers and claimholders are: a) the choice of effort by the managers, and b) the inability of managers to diversify away risk because their wealth is disproportionately allocated to the firm in the form of human capital investment.

⁶ Typical projects that may be avoided are those with long incubation periods, i.e. projects with prospects of poor near-term cash flow, but which might have excellent cash flow prospects beyond the manager's expected tenure. For example, returns from R&D projects often require considerable time in order to materialize and therefore may not facilitate managerial short-term goals (Laverty, 1996). Also, Dechow and Sloan (1991) provide empirical evidence that opportunistic reductions in R&D spending become more likely when CEOs approach retirement, i.e. when the “horizon” problem is most severe.

⁷ For example, as mentioned in Francis et al. (2005) and Easley and O'Hara (2004) better quality of accounting information can reduce the cost of capital by decreasing the (information-based) systematic risk of shares to uninformed investors.

⁸ In essence, this argument implies that the privately informed investors are better able to shift the portfolio weights to take advantage of any new information. Also, the results show support of the notion that in the presence of myopic CEOs, forecasting near-term cash flows becomes harder, thereby giving informed investors an additional edge over their uninformed counterparts.

2. Literature review and hypothesis development

Jensen and Meckling (1979) demonstrated that a shorter tenure increases the hurdle rate for projects causing managers to underinvest. Myopic managers prefer to make investments that offer relatively faster paybacks in order to enhance their reputations in the managerial labor market (see Campbell and Marino, 1994) or because they are preoccupied with safety⁹ (Hirshleifer and Thakor, 1992). This managerial “horizon problem” deems long-term investments that can increase firm value suboptimal from the perspective of managers.¹⁰

Fama (1980) argued that the managerial labor market can alleviate the agency problems between shareholders and senior executives. Accordingly, one can argue that the managerial labor market may provide discipline when managers' decision horizon is short. Brickley et al. (1999) provide some empirical evidence in support of this view. They show that performance in the CEO's final years influences the number of board seats he/she will hold as an outside director in other companies' boards after his/her retirement from the CEO position.¹¹ Davidson et al. (2007) argue that the labor market as a disciplinary mechanism is more effective in the case of younger executives at the beginning of their careers. We adopt this view and posit that concerns about their job market value may motivate managers in the early years of their careers to serve shareholder interests. However, as executives approach retirement, these career concerns may become either irrelevant, or less important than incentives provided by current compensation (Gibbons and Murphy, 1992). Therefore, as senior executives approach retirement, the potential for agency conflicts associated with the horizon problem becomes more severe. Consistent with this view, past studies have documented that long-tenured CEOs are more inclined to favor the status quo, in the sense that they are less likely to engage in new initiatives (Hambrick and Fukutomi, 1991).

The many arguments in favor of the notion that a firm stands to benefit from CEOs' expected future long-term tenure can also be motivated by the negative aspects associated with CEO longevity, i.e. the amount of years a CEO has already served in the firm.¹² Hambrick et al. (1993) show that firm and industry tenure are significantly associated with commitment to the status quo. Long-tenured CEOs grow stale in their jobs and ignore their environment, which impairs financial performance. Managers might also want to increase their expected tenure by initiating governance changes such as poison pills.¹³ It is important to note that past tenure should be differentiated from and is inversely related to *expected* (future) tenure. It makes sense that long-tenured CEOs could be set in their ways and might ignore competitive pressures that threaten a business model that has worked for a long time. However, long expected tenure should be a requisite for implementing a strategy, usually long-term in nature that would respond adequately to a dynamic environment. A CEO with short-term horizon might not make decisions that are in the best long-term interests of the firm. For example, Jensen (2004) has described how short-term managerial goals, often supported by earnings management and smoothing which involve the adoption of aggressive accrual accounting policies such as the premature recognition of revenues and earnings, can serve as potential sources of overvalued equity agency costs that end up destroying firm value. A CEO with a decidedly short-term focus could boost short-term profits by cutting costs, which is not a sustainable source of profit growth, rather than adding value by investing in positive NPV projects that do not generate immediate rewards.¹⁴ Thus, our first hypothesis is:

H1. Shorter CEO decision horizons are associated with greater agency costs.

Brickley et al. (1999) find that abnormal stock returns in the final two years of a CEO's tenure are particularly important in explaining the CEO's continued service on his own board. This conclusion implies that the prospect of continued board service can reduce horizon problems in the final years of a CEO's tenure. However a two-year span of stock price performance is a relatively short horizon, which to some extent, could induce myopia. For example, in an attempt to secure continued service on the company's board, the CEO could focus on appreciating the stock price by managing near-term earnings, thereby impeding accruals quality. Davidson et al. (2007) hypothesize that CEOs will be motivated to manage earnings prior to a turnover decision. This motivation comes from the aforementioned horizon problem, which is more severe for CEOs nearing retirement age. They find that firms in which CEOs are nearing retirement age have large discretionary accruals in the year prior to turnover. This is consistent with the notion that CEOs nearing retirement are not very concerned with either future earnings or the reversal in earnings that would follow earnings management to increase profits in the final years of their tenure. We therefore argue that information risk, as manifested in the quality of accruals (Francis et al., 2005),¹⁵ should be negatively related to expected tenure.

⁹ Chakraborty et al. (2007) determined that managers facing high termination risk make less risky investments, which might entail faster paybacks, than managers facing low termination risk.

¹⁰ Lavy (1996) looks at several determinants of economic “short-termism.” These are flawed management practice, managerial opportunism, stock market myopia, fluid and impatient capital, and information asymmetry. The one that is more relevant to our study is managerial opportunism.

¹¹ According to Fich and White (2005), reciprocal interlocking of CEOs, as manifested by representation on each other's boards, primarily benefits the CEOs' private interests rather than their shareholders.

¹² Bhagat and Bolton (2008) use the ratio of CEO tenure to CEO age as a measure of CEO quality, because a CEO who has had five years of tenure at age 65 is likely to be of different quality than a CEO that has had five years of tenure at age 50.

¹³ This argument has not been extensively tested in the literature. Nelson (2005) finds no relationship between governance changes, and CEO age, tenure or compensation.

¹⁴ It could be stated that the dissonance between DH and stakeholder time horizon is irrelevant because the stakeholder could always sell an underperforming stock. We would argue otherwise. The high execution costs of portfolio turnover for large institutional investors and the potential tax liabilities for investors would imply that investors would prefer a CEO with a long DH.

¹⁵ Accruals quality, as explained in Francis et al. (2005), tells investors about the mapping of accounting earnings into cash flows. Poor accruals quality weakens the mapping and increases information risk.

This is important because based on evidence from Easley and O'Hara (2004) and O'Hara (2003) firm-specific information risk, i.e. the possibility that firm-specific information used by investors in pricing stocks is of poor quality, is a priced risk. Thus, our second hypothesis is:

H2. Shorter CEO decision horizons should be associated with greater information risk.

Francis et al. (2004) show that accounting based earnings attributes capturing information quality, and in particular accruals quality, have a strong effect on firms' cost of capital. Thus, if the horizon problem affects the firms cost of capital it should also have an impact on market valuation. Moreover, as we have argued before, in the absence of the horizon problem, long expected tenure creates an environment that encourages CEOs to focus on the long-term interests of shareholders. Because shareholders have a long-term focus, and if the rewards reaped from turnarounds and other corporate strategies are expected to occur in the long-term, it stands to reason that long-term CEO tenure is in the best interest of shareholders. In contrast, CEOs with short-term expected tenure may display “myopic” behavior, i.e. focus on near-term quick fixes that focus on stock appreciation or wealth maximization in the immediate future. Thus, our third hypothesis is formulated as follows:

H3. Longer CEO decision horizons are associated with greater market valuations.

3. Data sources and measure of CEO's decision horizon

3.1. Data

We collected CEO's career information and compensation data from the sample of firms in Standard and Poor's (S&P) ExecuComp database. ExecuComp covers the period from years 1992 to 2003, and includes data for firms in the S&P 1500 index, which comprises the S&P 500, the S&P 400 mid cap, and the S&P 600 small cap indices. ExecuComp also contains information on firms that are not currently in the S&P500, the S&P400, and the S&P 600 indices, but were previously included in one of the aforementioned indices. Thus, ExecuComp includes data over 12 years with CEO characteristics and compensation information, which is equivalent to about 1500 observations per year. We require that the sample firms have information on CEO age and tenure in order to compute the “expected tenure” measure, which is our proxy for CEO horizon and a key variable in our tests. Thus, in order for us to be able to obtain the expected length of CEOs' tenure, the sample firms are required to have information available on the age, the commencement date and termination date of their CEO. According to ExecuComp, CEOs' total compensation is comprised of seven categorized items: 1) salary, 2) bonus, 3) options granted, 4) restricted stocks granted, 5) long-term incentive plan, 6) other annual compensation, and 7) all other compensation. We exclude from our sample firms that have missing information on any of the above seven categories. There are 2389 observations with complete information on the compensation package components and the items necessary to compute the expected tenure measure (i.e., tenure and age).¹⁶ We also omitted all observations from the first four years, 1992 to 1995, because the data availability for these years is quite insufficient with only about 30% of usable data compared to other years. Therefore, we restrict the sample period to the most recent 8 years with available data, i.e. the period from 1996 to 2003.¹⁷

Financial data are collected from *Compustat*, and information related to stock price is extracted from the *Center for Research in Securities Prices (CRSP)* database. Our sample is selected from the ExecuComp dataset, if accounting and financial information is available in the *Compustat* and *CRSP* databases. We are able to collect 1135 observations, which include all information needed to test our hypotheses.

3.2. Measure of CEO's decision horizon

We start our empirical investigation from the premise that CEO decision horizon can be adequately approximated by his/her expected tenure with the firm (Jensen and Meckling, 1979). We also assume that CEOs estimate their tenure by comparing themselves with other CEOs in their industry. The comparison with the other CEOs is done in two dimensions: the length of current tenure and age. Consequently, our measure of CEO decision horizon (DH) is constructed as follows.

$$DH_{i,t} = [TENURE_{ind,t} - TENURE_{i,t}] + [AGE_{ind,t} - AGE_{i,t}] \quad (1)$$

where $TENURE_{i,t}$ is the number of years the CEO has held that position and $AGE_{i,t}$ is the age of the CEO who work for firm i in year t . $TENURE_{ind,t}$ and $(AGE_{ind,t})$ is the industry median of $TENURE$ (AGE).¹⁸ Industry is defined based on the Fama-French 12 industry classification.

Given that the DH measure is an industry-adjusted measure, it can take either positive or negative values. A positive value indicates that the CEO's expected tenure is longer than the industry median because the CEO is younger and/or has not been in his/her current position as long as the industry median. Similarly, a negative value indicates that the expected tenure is shorter than the industry median because the CEO is older and/or been in his/her position for longer time than the median competitor firm's CEO.

¹⁶ We filled missing information on CEO age by using the data available on *Forbes.com*. The detail information about CEO's age is available on the web site, <http://www.forbes.com>.

¹⁷ When we repeat the tests using the complete sample that includes the years 1992 to 1995 we obtain qualitatively similar results with those reported here.

¹⁸ In a series of robustness tests, described in Section 5.3, we also tested various models using alternative DH measures, including measures constructed with the industry mean values of AGE and $TENURE$ instead of the median values.

We acknowledge that the DH measure may be noisy. However, we believe that both relative age and relative tenure are important determinants of the true decision horizon, and valid proxies of myopia. Prior empirical studies examining the horizon problem proxied horizon by CEO age or by using an indicator variable that takes the value of one if the CEO is near retirement age.¹⁹ We contend that our measure is more complete because it does not only account for the impact of relative age on decision horizon but also accounts for relative tenure.²⁰ Moreover, we also acknowledge that the relationships we are examining may be endogenous in nature, and our empirical tests account for that.

It is noteworthy that, because the DH measure is an industry-adjusted measure, it is not a surrogate for the age of the industry. Industries in their infancy, such as internet related industries, would have companies with young management teams and young CEOs. Growth opportunities for these companies are a function of their technological innovation and the demand for their products and services, and not necessarily the ages of their CEOs, who typically tend to be younger than the CEOs' of mature companies such as General Motors and GE.

4. Empirical results: univariate tests

Table 1 presents descriptive statistics for all variables used in the paper. The average longevity, i.e. the number of years a CEO of our sample firms has held that position, is about six years. Consistent with findings in previous studies, the average CEO age is around 55 years. Because DH is computed as sum of tenure and age after adjusting for the industry median values, the median DH is zero and the average value is close to zero. The sample firms display a wide dispersion of CEO decision horizon. The standard deviation of DH is 8.2580, the 5th percentile value is –15, and 95th percentile value is 12. The average Tobin's q is 1.28. The mean total compensation of CEOs in our sample is about \$6 million, which comprise of salary, bonus, exercised options, restricted stock grants, long-term incentive plan, other annual compensation, and all other compensation. They hold unexercised stock options (OPTION) more than current total compensation (TCOMP). Mean ratio of OPTION to TCOMP is 1.5209. This is because unexercised stock options include all options granted before and in current year. In fact, ExecuComp provides two types of option values: exercised options and unexercised options. If options are already exercised by and paid to CEOs, the money from this option grant might be regarded as a regular salary. Consequently, it is reasonable to argue that the substantial amount of unexercised options that a CEO holds is more likely to induce him/her to create a problem related to CEO's myopia. The next set of variables is a collection of external governance measures, such as a corporate governance index (GI), analysts' coverage (AFC) and product market competition (CMPT). As shown in the mean value of the governance index (GI), our sample firms have 10 out of 28 anti-takeover provisions. On average, a firm in the sample is covered by 12 security analysts. The product markets of sample firms are quite competitive. The average book value of total assets is 10,569 million dollars, while long-term debt accounts for 23.43% of total assets. The average firm in the sample spends 1.82% and 1.13% of total sales revenues on R&D and advertising, respectively.

In the first-stage of our empirical investigation, we employ univariate tests designed to illustrate whether the level of CEO decision horizon is correlated with firm performance. To conduct this test, we rank firms on their CEOs' DH every year and then classify them into quintile groups. We present the positive relation between the two measures in Fig. 1. The average values of Tobin's q for the bottom and the top quintile DH firms are 1.12 and 1.42, respectively. This difference is statistically significant at 1% level with a t -statistic of 2.92. The other test results (not illustrated in figure) show that firms whose CEOs have longer horizons are smaller, less levered and more profitable. It is notable that the result of the univariate test of the relationship between CEO decision horizon and firm variables (i.e., performance or other characteristics) should be regarded with caution because this test does not control for other factors. The result of univariate test is therefore only preliminary evidence. The multi-factor test result of the relationship is reported in the next table.

5. Empirical results: multi-factor regression analysis

5.1. Empirical model development

Next, we develop an empirical model that examines how CEO decision horizon is related with important firm characteristics: 1) agency costs measured by free cash flows with poor growth opportunities, 2) information risk measured by accruals-based and microstructure-based measures, and 3) firm performance measured by Tobin's q . In this test where individual firms' performance (or, alternatively, agency costs or information risk) can be regressed on DH and firm-specific characteristics, the general form of the regression equation would be:

$$\text{Firm Performance} = f(\text{DH and firm characteristics}) \quad (2)$$

A potential limitation of this testing procedure is that the relationship between firm performance and CEO decision horizon could be endogenous. There are two types of endogeneity that apply here. First, it is possible that superior performing firms (or,

¹⁹ For example, Cheng (2004) used an indicator variable that equals 1 if the CEO is at least 63 years old, and 0 otherwise. This definition reflects the fact that CEOs typically retire between the ages of 64 and 66 (Murphy, 1999). Dechow and Sloan (1991) and Murphy and Zimmerman (1993) use the two years preceding the CEO retirement as a measure of the horizon problem. Bryan et al. (2000) use CEO age in years as the measure of closeness to retirement or the degree of the horizon problem.

²⁰ As a robustness check, we also tried a measure of DH using only the industry-adjusted age. The results we obtained are in line with those based on the DH measure that accounts for both relative CEO age and relative CEO tenure. These results are available upon request.

Table 1

Summary statistics. This table documents summary statistics for variables used in the paper. TENURE = the number of years the CEO has held his/her position, AGE = CEO age, DH = decision horizon, TOBINQ = Tobin's q , AQ = accrual quality, PIN = the annual probability of information-based trading, AGENCY = Agency costs are computed as $(FCF/TA) \times GROWTH$, where FCF = free cash flow, TA = total assets, GROWTH = growth dummy that takes one if the firm's TOBINQ is less than one (indicating a poorly managed firm) and zero otherwise. CEO compensation variables are: TCOMP = total compensation, OPTION = unexercised stock options, ECOMP = the ratio of equity compensation (OPTION) to total compensation (TCOMP), GI = corporate governance index, AFC = the number of security analysts' forecasts, CMPT = product market competition, TA = total assets, LEV = long-term debt scaled by total assets, M/B = market-to-book ratio, SALES = Gross sales, EBIT = earnings before interest and taxes, CAPX = capital expenditure, R&D = research and development expenditure, ADV = advertising expenditure, FIRMAGE = the number of years since the stock inclusion in the CRSP database.

	Mean	Std. Dev.	5th Percentile	Median	95th Percentile
Decision horizon					
TENURE	5.8344	4.2709	1.0000	5.0000	14.000
AGE	55.352	5.7877	46.000	55.000	64.000
DH	−0.6722	8.2580	−15.000	0.0000	12.000
Firm performance					
TOBINQ	1.2761	1.1071	0.4294	0.9286	3.0403
Information risk					
AQ1	0.0199	0.0174	0.0041	0.0154	0.0462
AQ2	0.0172	0.0122	0.0037	0.0142	0.0377
AQ3	0.0182	0.0134	0.0040	0.0155	0.0398
AQ4	0.0306	0.0195	0.0078	0.0267	0.0674
PIN	0.1195	0.0352	0.0745	0.1131	0.1811
Agency costs					
AGENCY	0.0140	0.0278	0.0000	0.0000	0.0739
CEO compensation					
TCOMP	6145.5	8688.0	1049.1	3749.0	19,259
OPTION	10,879	33,868	0.0000	2942.4	47,174
ECOMP	1.5209	2.3746	0.0000	0.8586	5.4270
External governance					
GI	10.365	2.3893	6.0000	11.000	14.000
AFC	11.770	8.3138	0.0000	11.000	36.000
CMPT	0.8997	0.1039	0.7239	0.9307	0.9811
Other firm characteristics					
TA (million \$)	10,569	18,793	473.87	4192.6	37,675
LEV	0.2343	0.1261	0.0129	0.2333	0.4382
M/B	1.8169	1.3309	0.9281	1.4415	3.7858
SALES (million \$)	9025.1	15,948	510.11	4122.5	30,147
EBIT/SALES	0.1085	0.1014	0.0145	0.0987	0.2572
CAPX/SALES	0.0764	0.0944	0.0150	0.0489	0.2178
R&D/SALES	0.0182	0.0346	0.0000	0.0021	0.0890
ADV/SALES	0.0113	0.0308	0.0000	0.0000	0.0680
FIRMAGE	28.304	12.417	3.0000	34.000	41.000

alternatively, firms with low agency costs or low information risk) are better at finding and hiring CEOs who are younger and/or are expected to have longer tenure, and therefore a long-term decision horizon. Since our hypothesis is that longer expected tenure (i.e., longer decision horizon) enhances firm performance, it may be difficult to identify the cause and effect between

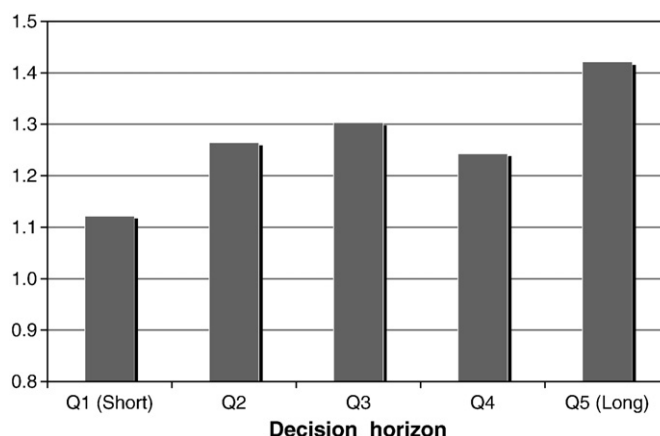


Fig. 1. CEO decision horizon and firm performance: univariate analysis. This figure presents the positive relation between CEO decision horizon (DH) and firm performance (TOBINQ). We rank firms on their CEOs' DH every year and then classify them into quintile groups. The average values of TOBINQ for the bottom and the top quintile DH firms are 1.12 and 1.42, respectively. This difference is statistically significant at 1% level.

expected tenure and firm valuation. If we fail to account for the possibility of an endogenous relationship between CEO decision horizon and firm performance by relying simply on an OLS model to test the relationship, the regression coefficients could be biased.²¹ Second, there may be some other important firm/CEO characteristics, which might be correlated with CEOs' expected tenure. Consider, for example, that CEO expected tenure is highly correlated with CEO compensation structure.²² If our model fails to control for the impact of compensation structure variables on expected tenure, the empirical results on firm value could be mistakenly attributed to expected tenure rather than to the real reason, which in this case would be the type of compensation. As we will show later, the ratio of unexercised stock options to total compensation (ECOMP) is significantly and negatively associated with DH.

To reduce the potential problem of endogeneity in the multi-factor regression models, we use a two-stage least squares (2SLS) model that involves the estimation of two regression models, one for CEO decision horizon and one for firm performance. The 2SLS procedure requires that the first-stage equation contain at least one instrumental variable that is unrelated to the error term in the second-stage model. Here, we use industry characteristics as instruments. Our full 2SLS model is structured as follows.

$$DH = f(\text{firm performance, firm characteristics, CEO compensation package, and instrumentals}), \quad (3)$$

where firm performance (TOBINQ) is computed as [market value of common equity + preferred stock liquidating value + long-term debt – (short-term assets – short-term liabilities)] / (total assets), following [Chung and Pruitt \(1994\)](#). Firm characteristics are: the log of total assets (SIZE), the ratio of long-term debt to total assets (LEV), profitability (EBIT/SALES), capital expenditure (CAPX/SALES), and intangibles' intensity that is measured as the sum of R&D expenditure (R&D) and advertising expenditure (ADV) scaled by SALES. The first-stage model also accounts for the fact CEOs approaching retirement are likely to receive different pay packages than those recently put into office (see [Anderson et al., 2006](#)).²³ For example, to mitigate the horizon problem, associated with CEOs nearing retirement, firms are likely to increase incentive-based compensation.²⁴ To account for that, we include a CEO's equity compensation ratio (ECOMP), which is computed as the value of unexercised stock options (OPTION) divided by the value of total compensation (TCOMP). [Gibbons and Murphy \(1992\)](#) argue that since career concerns are weaker when executives near retirement, incentive contracts should be the strongest for these workers. We therefore expect to find a negative relationship between DH and ECOMP. In addition, we include industry characteristics because the job market environment in the industry in which a CEO competes can be related to the CEO's decision horizon. Based on this expectation, we control for two variables: the percentage of other firms that are larger than the firm in the same industry (ISIZE) and the percentage of other CEOs who are paid more than the CEO in the same industry (ICOMP). Both ISIZE and ICOMP serve as proxies for the existence of better employment opportunities in the industry, which are expected to have an impact on CEO decision horizon. Specifically, we anticipate that in the presence of good employment opportunities in the industry, a CEO will be more inclined to adopt a long-term horizon in order to improve external employment opportunities in the future. Finally, several indicator variables that account for year, industry and exchange listing effects are also included. The industry indicators are based on the Fama-French 12 industry classification.

After, CEOs' decision horizon is estimated in the first-stage model, its fitted value is used as an independent variable in the second-stage models to explain firm performance, agency costs, or information risk.

$$\text{Firm performance} = f(\text{DH}^{\text{est}} \text{ and firm characteristics}) \quad (4)$$

5.2. Regression results

5.2.1. The determinants of CEO decision horizon

[Table 2](#) reports the estimates of coefficients of the first-stage regression model that estimates CEO decision horizon (DH). One of the two possible reasons for the existence of an endogenous relationship described in the previous section, was that superior performing firms (i.e., firm with high Tobin's q) may tend to hire CEOs who are more long-term oriented in their decision making process. The first-stage regression results support this notion. The coefficient of TOBINQ is positive and statistically significant, suggesting that superior performing firms are more likely to choose CEOs who are young and/or are expected to hold their CEO positions longer than other CEOs in their industry. In addition, as shown in the findings of the second-stage model, CEO decision horizon has a significantly positive impact on firm performance. The results indicate that there is reciprocal causality between CEO horizon and firm performance. Our second concern was that other variables could be highly correlated with expected tenure. Indeed, we find that decision horizon is negatively related with the ratio of unexercised stock options to total compensation

²¹ Following a referee's suggestion, we attempt to determine causality using a simple cross-lagged regression model following [Davidson et al. \(1997\)](#). We control for the lagged variables of both decision horizon (DH) and Tobin's q in the separate regressions of DH and Tobin's q . We find that both lagged variables are significantly related with Tobin's q and DH, indicating an endogenous relation between the two variables. The regression results are available upon request.

²² [Brookman and Thistle \(2009\)](#) determined that CEO compensation from salary, bonus and long-term incentive plans are positively related to expected tenure but that option compensation is endogenous and unrelated to CEO tenure. They also found that greater CEO stock ownership increases expected tenure.

²³ [Schotter and Weigelt \(1992\)](#) also used an experimental setting and found that subjects' discounting of future outcomes was dependent upon compensation schemes.

²⁴ However, neither [Eaton and Rosen \(1983\)](#) nor [Yermack \(1995\)](#) document a significantly positive relationship. By contrast, [Lewellen et al. \(1987\)](#) do find a significantly positive relationship.

Table 2

The determinants of CEO decision horizon. This table reports coefficients from estimating the first-stage regression of the 2SLS model. The first step involves a regression model wherein decision horizon (DH) is estimated. The predicted value of DH is used in the second-stage models in other tables. DH = decision horizon, SIZE = log of total assets, LEV = long-term debt scaled by total assets, EBIT = earnings before interest and taxes, CAPX = capital expenditure, R&D = research and development expenditure, ADV = advertising expenditure, SALES = Gross sales, TOBINQ = Tobin's *q*, ISIZE = the percentage of other firms that are larger than the firm in the same industry, ICOMP = the percentage of other CEOs who are paid more than the CEO in the same industry, ECOMP = the ratio of equity compensation to total compensation. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

	Dependent variable = Decision horizon (DH)	
	Estimated coefficient	t-statistic
INTERCEPT	0.6359	0.06
SIZE	−0.1223	−0.25
LEV	−3.7354*	−1.68
EBIT/SALES	−2.8348	−1.05
CAPX/SALES	−1.7314	−0.58
R&D/SALES	−5.6253	−0.50
ADV/SALES	17.2455*	1.92
TOBINQ	0.7980**	2.55
ISIZE	0.7011	0.37
ICOMP	2.6801***	2.68
ECOMP	−0.4948***	−4.42
Year dummies	Yes	
Industry dummies	Yes	
Exchange dummies	Yes	
N	1135	
R ²	12.78%	
F [Prob. >F]	5.21*** [0.0000]	

(ECOMP). The negative estimated coefficient of ECOMP is consistent with the evidence in [Gibbons and Murphy \(1992\)](#), and indicates that a CEO whose decision horizon is short normally has substantially greater amount of options that are not exercised, which can lead to a short-term tenure-driven myopia. This evidence provides an explanation for the reason why some compensation structures that are not optimally designed may negatively affect firm performance by increasing information risk and lowering firm valuation through a decision horizon channel. Another interesting result from the first-stage is that the coefficient of ICOMP is significantly positive. This is consistent with the notion that younger CEOs with relatively few years of tenure that have longer horizons are normally paid less than the other CEOs in the same industry.

5.2.2. CEO decision horizon and agency costs

Our first hypothesis states that longer CEO decision horizons should be negatively associated with firms' agency costs. To conduct a test of this hypothesis, we devise a measure of agency costs of free cash flows based on the notion that if a firm has cash flows in excess of that needed to fund all positive NPV projects, then there is a greater chance that managers may use the free cash flows to invest in negative NPV projects that increase their personal utility.²⁵ The agency conflicts involving free cash flows are likely to be prevalent in low growth firms because they generally have substantial free cash flow, which managers could decide to overinvest. In contrast, high growth firms are not as likely to suffer from the free cash flow agency problem because they have an abundance of good growth opportunities. These firms are usually short of cash after using internal funds for funding new projects and often need to rely on external financing to cover their financing needs. Following [Doukas et al. \(2000\)](#), we proxy agency costs of free cash flow using the interaction of a poor growth opportunities indicator with free cash flows standardized by total assets.

$$\text{AGENCY}_{i,t} = \left(\frac{\text{FCF}_{i,t}}{\text{TA}_{i,t}} \right) \times \text{GROWTH}_{i,t}, \quad (5)$$

where free cash flow (FCF) is measured as operating income before depreciation minus the sum of taxes, interest expense, and dividends paid (see [Lehn and Poulsen, 1989](#)). FCF is standardized by total assets (TA). GROWTH is an indicator variable that takes the value of 1 if the firm's Tobin's *q* is less than 1 (indicating a poorly managed firm or a firm with poor growth opportunities) and the value of 0 otherwise.

In the second stage of our analysis wherein we examine the determinants of agency costs, we include as independent variable the fitted value of the decision horizon from the first-stage regression shown in [Table 2](#), along with a set of control variables chosen based on models in previous studies (e.g., [Bergstresser and Philippon, 2006](#)). The control variables are external governance variables, and other firm characteristics. [Table 3](#) provides two regression results. In Column [1], in addition to the fitted value of CEO decision horizon, we control for other firm characteristics like size, leverage and firm age, that may help explain firm's agency

²⁵ This draws from [Jensen's \(1986\)](#) free cash flow theory, where it is argued that debt can be used to impose discipline on managers in firms with large free cash flows.

Table 3

CEO decision horizon and agency costs. This table shows the 2SLS regressions of agency costs on CEO's decision horizon and reports the estimates of coefficients of the second-stage regressions. The first-stage model estimates decision horizon (DH) in Table 3. Agency costs (AGENCY) are measured as $(FCF/TA) \times GROWTH$, where FCF = free cash flow, TA = total assets, GROWTH = growth dummy that takes one if the firm's TOBINQ is less than one (indicating a poorly managed firm) and zero otherwise. The first-stage regression estimates DH in Table 2. DH = decision horizon, GI = corporate governance index, AFC = the number of security analysts' forecasts, CMPT = product market competition, SIZE = log of total assets, LEV = long-term debt scaled by total assets, FIRMAGE = the number of years since the stock inclusion in the CRSP database, M/B = market-to-book ratio, The Durbin–Wu–Hausman (DWH) test is employed to test the endogenous relation, and Chi-squared and p-value are provided. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

	Dependent variable = Agency costs (AGENCY)	
	[1]	[2]
INTERCEPT	0.0797*** (4.67)	0.0572*** (2.80)
DH	−0.0014*** (−3.02)	−0.0014*** (−2.94)
External governance		
7 ≤ GI ≤ 9 (democracy)		0.0015 (0.38)
10 ≤ GI ≤ 12		0.0019 (0.47)
13 ≤ GI (dictatorship)		0.0032 (0.76)
LN(1 + AFC)		−0.0027* (−1.84)
CMPT		0.0072 (0.87)
Other firm characteristics		
SIZE	−0.0035*** (−4.73)	−0.0025*** (−2.71)
LEV	−0.0156** (−2.08)	−0.0175** (−2.32)
LN(1 + FIRMAGE)	0.0004 (0.35)	0.0004 (0.29)
10 deciles of M/B	Yes	Yes
Year dummies	Yes	Yes
Exchange dummies	Yes	Yes
DWH test	5.87** [0.0154]	5.52** [0.0188]
N	989	989
R ²	15.45%	15.83%
F	7.34***	6.96***
[Prob. >F]	[0.0000]	[0.0000]

costs. We find that the coefficient of CEO decision horizon is −0.0014 and it is statistically significant at 1%. This suggests that a one standard deviation increase in CEO decision horizon is associated with a 1.2% decrease in agency costs. The leverage ratio is significantly and negatively related to agency costs, which lends support to Jensen's (1986) free cash flow theory, and is consistent with the notion that debt can be used to alleviate agency costs of free cash flows. In Column [2], we re-estimate the model after adding additional controls in the form of external governance variables. These variables are a series of indicator variables based on the governance index of Gompers et al. (2003) (GI dummies), the number of analyst forecasts (AFC), and product market competition (CMPT). The GI dummy variables are based on the corporate governance index (GI) of Gompers et al. (2003), which proxies for the level of shareholder rights.²⁶ Based on Jensen's (1986) argument that the threat of takeover is a strong mechanism of managerial discipline, a firm with a high governance index (i.e., many anti-takeover provisions) is expected to have a high level of agency problems. Second, security analysis can act as a monitoring mechanism in reducing agency costs (Doukas et al., 2000), and therefore, it is expected that analysts' coverage is associated with the lower agency costs. Analysts' coverage (AFC) is measured as the number of security analysts' one fiscal year-ahead forecasts collected every June from the I/B/E/S Detail History Database. Third, product market competition can be regarded as an external governance mechanism because the competition in the product markets drives prices towards minimum average cost, thereby motivating managers to increase firm efficiency. Our proxy for the competition in the product market is computed as the inverse value of Herfindahl concentration index.

$$CMPT_{i,t} = 1 - \text{Herfindahl concentration index} = 1 - \left[\frac{\sum_j (\text{SALES}_{j,t})^2}{\left(\sum_j \text{SALES}_{j,t} \right)^2} \right] \quad (6)$$

²⁶ GI is constructed by counting the existence of 28 different provisions related to shareholder protection and listed in 5 categories: Delay, Protection, Voting, Other, and State. Among the 28 provisions, 24 are unique and enter the index with equal weight. Gompers et al. (2003) construct the governance index without requiring any judgment about the efficacy or wealth effects of any of these provisions, but consider their impact on the balance of power between managers and outside shareholders.

where $SALES_{j,t}$ is the annual sales of firm j belonging to the industry in which firm i is included in year t . If the total amount of sales in the industry is dominated by few firms, then the Herfindahl index will show a high value that is close to one. Higher values of CMPT (i.e., lower Herfindahl index) thus indicate a more competitive product market and low agency costs. The results obtained after adding these control variables, shown in column [2], remain quantitatively and qualitatively similar to those in column [1]. The only external governance variable that shows a significant effect on agency costs is analyst following, which enters the regression with a negative coefficient, indicating that analyst coverage acts as a monitoring mechanism that reduces agency costs.

The Durbin–Wu–Hausman test employed to identify the endogenous relation is reported on the bottom of the table. The test is based on the premise that the OLS estimator is more efficient than the 2SLS estimator in the absence of endogeneity. We find that the test results in Table 3 and the following tables show significant values in chi-squared, justifying the use of 2SLS.

5.2.3. CEO decision horizon and information risk

Next, using the 2SLS procedure we investigate how decision horizon is related with the level of information risk investors face. Prior research has provided evidence that companies a) report lower earnings following a CEO succession (Wells, 2002), b) manage earnings following a CEO succession (Pourciau, 1993), and c) positively manage earnings prior to CEO turnover (Davidson et al., 2007). Thus, given that CEO turnover is indicative of short decision horizon, we hypothesize that both accruals- and microstructure-based measures of information risk should be associated with shorter CEO decision horizons. Investigating the impact of decision horizon on accruals is motivated by Jensen and Meckling (1979) and the subsequent empirical evidence of Francis et al. (2004, 2005) that poor accruals quality (one proxy for information risk) causes high costs of debt and equity and thus leads to lower firm value. Information risk arises when some investors have more and better information than others about a firm's prospects. According to Aslan et al. (2007), the size of such risk depends on a variety of factors such as the quality of a firm's accounting information, the availability of public information sources about the firm, the frequency of new information events, and the fraction of investors who have better information.

Drawing from both the accounting and finance literatures, we construct five alternative measures of firm's information risk. The first four measures capture information risk related to the “uncertainty” of firm's performance by estimating the quality of accruals. The last measure of information risk, the probability of information-based trading (PIN), is a market microstructure measure that captures the level of information risk between investors. The first measure of accruals quality (AQ1) is computed as the standard deviation of firm-specific residuals from a regression of total accruals on lagged, contemporaneous, and leading cash flow from operations as defined in Dechow and Dichev (2002).

$$\frac{TCA_{i,t}}{TA_{i,t}} = k_0 + k_1 \frac{CFO_{i,t-1}}{TA_{i,t}} + k_2 \frac{CFO_{i,t}}{TA_{i,t}} + k_3 \frac{CFO_{i,t+1}}{TA_{i,t}} + \varepsilon_{i,t}, \quad (7)$$

where $TCA_{i,t} = (\Delta CA_{i,t} - \Delta CL_{i,t} - \Delta CASH_{i,t} + \Delta STDEBT_{i,t} - DEPN_{i,t})$ = firm i 's total current accruals in year t , $\Delta CA_{i,t}$ = change in current assets between year $t-1$ and year t , $\Delta CL_{i,t}$ = change in current liabilities between year $t-1$ and year t , $\Delta CASH_{i,t}$ = change in cash between year $t-1$ and year t , $\Delta STDEBT_{i,t}$ = change in debt in current liabilities between year $t-1$ and t , $DEPN_{i,t}$ = depreciation and amortization expense in year t , $CFO_{i,t}$ = firm i 's cash flow from operations in year t that is given in the Statement of Cash Flows reported under the Statement of Financial Accounting Standards No. 95 (SFAS No. 95, FASB 1987). All variables are scaled by average total assets ($TA_{i,t}$), and the regression is based on 10-year rolling windows.

The second measure of accruals quality (AQ2) is constructed in a similar manner as the first one, except that cash flow from operations (CFO) is computed as net income before extraordinary items minus total accruals, rather than using CFO reported under SFAS No. 95.

To obtain the third measure of accruals quality (AQ3), we regress the change in working capital on lagged, contemporaneous, and leading cash flow from operations, and compute the standard deviation of firm-specific residuals from the regression.

$$\frac{\Delta WC_{i,t}}{TA_{i,t}} = k_0 + k_1 \frac{CFO_{i,t-1}}{TA_{i,t}} + k_2 \frac{CFO_{i,t}}{TA_{i,t}} + k_3 \frac{CFO_{i,t+1}}{TA_{i,t}} + \varepsilon_{i,t}, \quad (8)$$

where the dependent variable for this regression is the change in working capital between year $t-1$ and t , $\Delta WC_{i,t} = \Delta AR_{i,t} + \Delta INV_{i,t} + \Delta AP_{i,t} - \Delta TP_{i,t} + \Delta OA_{i,t}$, $\Delta AR_{i,t}$ = change in accounts receivable, $\Delta INV_{i,t}$ = change in inventory, $\Delta AP_{i,t}$ = change in accounts payable, $\Delta TP_{i,t}$ = change in taxes payable, and $\Delta OA_{i,t}$ = change in other assets.

Finally, we create the fourth measure of accruals quality (AQ4) following Francis et al. (2005) wherein Eq. (9) below is estimated for each of Fama and French's (1997) 48 industry groups. This procedure is used to capture the industry effect on firm's accrual quality. We compute the standard deviation of firm-specific residuals from the cross-sectional regression for each of the 48 industry groups.

$$\frac{TCA_{i,t}}{TA_{i,t}} = k_0 + k_1 \frac{CFO_{i,t-1}}{TA_{i,t}} + k_2 \frac{CFO_{i,t}}{TA_{i,t}} + k_3 \frac{CFO_{i,t+1}}{TA_{i,t}} + k_4 \frac{\Delta SALES_i}{TA_{i,t}} + k_5 \frac{PPE_{i,t}}{TA_{i,t}} + \varepsilon_{i,t}, \quad (9)$$

where the regression controls for sales change and PPE in addition to the lagged, contemporaneous, and leading cash flows. $\Delta SALES_i$ = change in sales between year $t-1$ and t , and $PPE_{i,t}$ = property, plant, and equipment in year t .

Table 4

CEO decision horizon and information risk. This table shows the 2SLS regressions of information risk on CEO's decision horizon and reports the estimates of coefficients of the second-stage regressions. The first-stage model estimates decision horizon (DH) in Table 2. The level of information risk is measured as accrual quality (AQ) and the probability of information-based trading (PIN). DH = decision horizon, GI = corporate governance index, AFC = the number of security analysts' forecasts, CMPT = product market competition, SIZE = log of total assets, LEV = long-term debt scaled by total assets, FIRMAGE = the number of years since the stock inclusion in the CRSP database, M/B = market-to-book ratio. The Durbin–Wu–Hausman (DWH) test is employed to test the endogenous relation, and Chi-squared and *p*-value are provided. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Dependent variable	Accrual quality 1	Accrual quality 2	Accrual quality 3	Accrual quality 4	PIN	Ln(1 + PIN)
	[1]	[2]	[3]	[4]	[5]	[6]
INTERCEPT	0.1229*** (9.94)	0.0908*** (10.48)	0.0951*** (9.07)	0.1218*** (9.04)	0.5063*** (17.33)	0.4546*** (17.79)
DH	−0.0008*** (−2.63)	−0.0007*** (−3.60)	−0.0006** (−2.51)	−0.0007** (−2.37)	−0.0011* (−1.92)	−0.0010* (−1.90)
External governance						
7 ≤ GI ≤ 9 (democracy)	−0.0004 (−0.17)	−0.0002 (−0.11)	−0.0012 (−0.58)	0.0044* (1.68)	−0.0005 (−0.19)	−0.0003 (−0.08)
10 ≤ GI ≤ 12	−0.0039 (−1.58)	−0.0024 (−1.36)	−0.0005 (−0.26)	0.0046* (1.79)	−0.0030 (−0.67)	−0.0025 (−0.65)
13 ≤ GI (dictatorship)	−0.0031 (−1.19)	−0.0005 (−0.25)	−0.00004 (−0.02)	0.0072** (2.57)	−0.0005 (−0.10)	−0.0003 (−0.08)
LN(1 + AFC)	0.0006 (0.72)	0.0004 (0.72)	−0.0007 (−0.93)	0.0003 (0.33)	−0.0061*** (−3.41)	−0.0052*** (−3.36)
CMPT	−0.0086* (−1.70)	−0.0082** (−2.31)	−0.0033 (−0.80)	−0.0175*** (−2.82)	−0.0087 (−0.81)	−0.0071 (−0.76)
Other firm characteristics						
SIZE	−0.0038*** (−6.78)	−0.0024*** (−6.20)	−0.0027*** (−5.56)	−0.0024*** (−3.90)	−0.0150*** (−12.73)	−0.0133*** (−12.87)
LEV	0.0007 (0.16)	−0.0084 (−2.61)	−0.0160** (−3.99)	−0.0159*** (−3.24)	−0.0103 (−1.12)	−0.0087 (−1.07)
LN(1 + FIRMAGE)	−0.0007 (−0.84)	−0.0004 (−0.33)	−0.0023*** (−2.84)	−0.0074*** (−8.74)	−0.0010 (−0.65)	−0.0009 (−0.72)
10 deciles of M/B	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Exchange dummies	Yes	Yes	Yes	Yes	Yes	Yes
DWH test	1.86 [0.1727]	8.56*** [0.0034]	20.19*** [0.0000]	1.92 [0.1654]	6.95*** [0.0084]	6.76*** [0.0093]
N	1004	972	862	1011	671	671
R ²	22.34%	23.01%	18.17%	21.29%	45.18%	45.71%
F	10.02***	10.07***	6.61***	9.49***	20.41***	20.85***
[Prob. > F]	[0.0000]	[0.0000]	[0.0000]	[0.0000]	[0.0000]	[0.0000]

Recent research has also utilized the probability of information-based trading (PIN) to proxy private information risk (i.e., information asymmetry between investors). According to Easley et al. (1996a, 1996b, 1997a, 1997b, 2002, 2005), the PIN is estimated as the ratio of expected informed order flow to total order flow:

$$PIN = \frac{\alpha\mu}{\alpha\mu + \varepsilon_s + \varepsilon_b}, \quad (10)$$

where α is the probability that an information event occurs, μ is the arrival rate of informed trades, and ε_s and ε_b are the arrival rates of uninformed sells and buys respectively. $\alpha\mu$ is the expected arrive rate of informed trades and $\alpha\mu + \varepsilon_s + \varepsilon_b$ is the arrival rate for all orders.²⁷ PIN is the fraction of orders that arise from informed traders or the probability that the opening trade is information-based.

The results of the second-stage regressions are reported in Table 4. Note that the fitted value of DH is used as an independent variable in the second-stage models. The results reveal that the estimated coefficient of CEO decision horizon is negative and significant at conventional levels for all measures of information risk (note that the lower numbers of accruals quality and PIN represent lower degree of information risk) even after controlling for external governance measures, size, leverage, market-to-book, firm age, exchange and industry effects. Some of the other variables also manifest significant associations with accrual quality. In particular, firm size is negatively associated with accrual quality, consistent with the evidence provided in Bergstresser and Philippon (2006). As expected our results show that product market competition motivates managers to mitigate agency costs, which in turn reduces information risk.

²⁷ The yearly estimates of probability of information-based trading (PIN) are available on Soeren Hvidkjaer's web site: <http://www.smith.umd.edu/faculty/hvidkjaer/data.htm>.

Table 5

CEO decision horizon and firm performance. This table shows the 2SLS regressions of Tobin's q on CEO's decision horizon and reports the estimates of coefficients of the second-stage regressions. The first-stage model estimates decision horizon (DH) in Table 2. DH = decision horizon, SIZE = log of total assets, LEV = long-term debt scaled by total assets, EBIT = earnings before interest and taxes, CAPX = capital expenditure, R&D = research and development expenditure, ADV = advertising expenditure, SALES = Gross sales, TOBINQ = Tobin's q , ISIZE = the percentage of other firms that are larger than the firm in the same industry, ICOMP = the percentage of other CEOs who are paid more than the CEO in the same industry, ECOMP = the ratio of equity compensation to total compensation. The Durbin–Wu–Hausman (DWH) test is employed to test the endogenous relation, and Chi-squared and p -value are provided. * and *** denote significance at the 10% and 1% levels.

	Dependent variable = TOBINQ		
	[1]	[2]	[3]
INTERCEPT	1.7886*** (10.67)	−0.9476* (−1.89)	−0.5249 (−1.04)
DH	0.1152*** (7.76)	0.1590*** (9.38)	0.1562*** (8.80)
SIZE		0.1211*** (5.34)	0.0912*** (3.88)
LEV		−1.5546*** (−7.26)	−1.2385*** (−5.77)
EBIT / SALES		3.1794*** (12.64)	3.1058*** (12.55)
CAPX / SALES		2.215*** (7.69)	2.0685*** (7.31)
R&D / SALES			7.3759*** (7.01)
ADV / SALES			0.1562*** (8.80)
Year dummies	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes
Exchange dummies	Yes	Yes	Yes
DWH test	54.60*** [0.0000]	79.34*** [0.0000]	70.61*** [0.0000]
N	1135	1135	1135
R ²	40.98%	53.58%	55.56%
F	35.10***	49.19***	49.39***
[Prob. >F]	[0.0000]	[0.0000]	[0.0000]

5.2.4. CEO decision horizon and market valuation

Since we have shown that CEO decision horizon is negatively associated with agency costs and information risk, it is reasonable to expect that firms with long CEO decision horizon show superior market value-based firm performance. Accordingly, our next test examines the link between decision horizon and performance. We report the estimates of coefficients for the 2SLS models of decision horizon on market value-based performance (Tobin's q) in Table 5. The results of second-stage models in Columns [1] to [3] confirm the prediction of our main hypothesis that there is a positive impact of CEO decision horizon on firm value. After controlling for other firm variables (see Column [2]), we find that the coefficient of CEO horizon is 0.159 with a t -statistic of 7.76. This suggests that a ten-year difference, for example, between median industry CEO age (or tenure) and CEO age (i.e., when the DH variable takes the value of ten) generally results in an increase in Tobin's q by 1.59 points. Given the fact that the average Tobin's q is 1.28 with a standard deviation of 1.11, the economic effect of decision horizon on firm value is also significant. In Column [3], we re-estimate the model after adding intangible intensity variables (R&D/SALES and ADV/SALES), which are also considered as important contributors to firm performance. The results are very similar to those obtained from regression [2]. Consistent with prior evidence (see Morck and Yeung, 1991, among others), we also find that the intangibles are significant contributors to firm value. The coefficients of the other control variables show the expected signs. Market valuation is positively related to size, profitability, and capital expenditure, while negatively related to leverage. By including industry dummy variables in all regressions, we are controlling for industry effects on market valuation.

5.3. Robustness tests

We performed several robustness tests to ensure that our findings are not due to the particular measures of decision horizon and firm performance, and are not driven by the specific estimation methodology used. Robustness tests were performed in the following areas: 1) alternative performance measures (earnings-per-share, and excess valuation measures based on Berger and Ofek (1995) and Rhodes-Kropf et al. (2005)), 2) alternative measures of decision horizon (e.g., using only the CEO's age or retirement age in the construction of DH), 3) alternative ways to estimate decision horizon (e.g., by accounting for cases where the CEO's tenure ended because of getting fired or because he was deceased, by considering the board's age, etc.), 4) alternative methods of estimation (e.g., using three-stage least squares regressions, or firm fixed effects regressions and CEO fixed effects regressions). We find that our conclusions are unchanged using alternative measures and methods of estimation. For the sake of brevity, we exclude the results of these tests from this version of the

paper. A more detailed version of this paper that presents all of the robustness tests is available from the authors upon request.²⁸

6. Summary and conclusions

The prior managerial horizon literature, presented in earlier sections of the paper, has linked managers' propensity to forgo long-run investments with their limited tenure. With average CEO tenure having decreased from about 8 to less than 4 years over the past 20 years, pressure on CEOs to deliver quick results has increased dramatically. One of the main sources of conflict between managers and firms' stakeholders is that managers' decision horizons are shorter than shareholders' investment horizons. Managers' claim on the firm is limited to their tenure, while a firm's lifespan is much longer. Since managerial decision horizons are limited to their expected tenure, managers approaching retirement age or about to be replaced become more *myopic*, in the sense that they tend to place less weight on cash flows occurring after their employment time horizon. A consequence of managerial myopia is that some good projects may be avoided. Conversely, the managerial labor market can alleviate the agency problems between shareholders and senior executives. It has been argued that the managerial labor market may provide discipline when managers' decision horizon is short. Performance in the CEO's final years influences the number of board seats he/she will hold as an outside director in other companies' boards after his/her retirement from the CEO position. Therefore, one could posit that post-retirement employment opportunities would prevent CEOs from making myopic decisions, which would make the decision horizon debate irrelevant. These questions are empirical in nature and can be subject to such investigation.

In this study we empirically test whether there is a link between CEO decision horizon and agency costs as well as firm's performance. In particular, we investigate how CEO expected tenure impacts financial performance, information risk, and agency costs of free cash flow. We hypothesize that short CEO decision horizons (CEO myopia or horizon problem) should be positively associated with agency costs and negatively related to firm performance. We find strong evidence that shorter CEO decision horizons are associated with significant agency costs. We also confirm the hypothesis that decision horizon has a positive influence on firm performance. We find that firms with CEOs that have long decision horizons have high market valuation, as measured by Tobin's *q* and other alternative measures of firm valuation. Our results also support the notion that decision horizon substantially mitigates the level of information risk that investors face. The evidence we provide in this paper is consistent with the notion that myopic managers prefer to make investments that offer relatively faster paybacks at the expense of long-term value creation.

Our results also imply that it is imperative for firms to design compensation structures that would discourage myopic behavior and conversely encourage CEOs to lengthen their decision horizon. Hiring a young CEO is not necessarily a sufficient solution to the problem. The potential to earn a huge windfall in the short-term could induce the CEO to make decisions that are detrimental to long-term shareholder wealth maximization. Stock options, originally intended to lengthen decision horizon, actually shorten the horizon, and thus may lead to inferior firm performance.

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